

Benjamin Knepper

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EDUCATION

University of California, Berkeley , Department of Physics	Berkeley, CA
Graduate Student Researcher, PhD Candidate	08/2026 - present
University of California, Berkeley , College of Letters & Science	Berkeley, CA
B.A. Physics (with honors), B.A. Philosophy, Minor in Mathematics	08/2023 - 12/2025
Physics GPA: 3.98/4.0	
Cornell University , College of Arts & Sciences	Ithaca, NY
Cumulative GPA: 3.77/4.0	08/2020 - 05/2022

PUBLICATIONS

<i>In Progress:</i>	1. P. Höhn, B. Knepper, L. Marchetti, <i>To Appear</i> (2026).
	2. M. Karydas, B. Knepper, D. Carney, J. Sonner, <i>To Appear</i> (2026).
<i>Peer Reviewed:</i>	3. G. Hoshino et al. (GigaBREAD Collaboration), Phys.Rev.Lett. 134 171002 (2025).
	4. M. Tala Pinto et al., Astronomy & Astrophysics 694 A268 (2025).
	5. S. Knirck et al. (BREAD Collaboration), Phys.Rev.Lett. 132 131004 (2024), <i>Editor's Suggestion</i> .
<i>Manuscripts:</i>	6. B. Knepper, A. Sonnenschein, S. Knirck, OSTI 2377355 (2024).

RESEARCH

Okinawa Institute of Science and Technology (OIST)	Okinawa, Japan
<i>Research Intern, Qubits and Spacetime Unit</i>	04/2026 - 07/2026
- Formulated a covariant quantization approach to perspective-neutral quantum reference frames via a relational, gauge-invariant path integral - Explicitly constructed this path integral for a mechanical model and generalizing to arbitrary gauge groups	
UC Berkeley Physics Honors Bachelor's Thesis	Berkeley, CA
<i>Student Researcher</i>	03/2025 - present
- Calculating the “standard quantum limit” of measuring the length of a Jackiw-Teitelboim gravity wormhole in the Hartle-Hawking state, using the gravitational path integral and input-output linear response theory - Devising both digital (qubit) and analog (cold atom cavity) experimental implementations	
Lawrence Berkeley National Laboratory	Berkeley, CA
<i>QuIPS (Quantum Invisible Particle Sensor) Experiment Researcher</i>	12/2023 - 08/2024
- Collaborating on Geant4 Monte Carlo simulations of unique first-forbidden β^- decays and analyses of signal-to-noise projections for the upcoming QuIPS beyond-standard-model search - Designed optimal configuration of particle physics calorimeters surrounding an optomechanically levitated nanosphere, achieving a QuIPS detector sensitivity to a sterile neutrino coupling of 10^{-3}	
Network for Neutrinos, Nuclear Astrophysics, and Symmetries (N3AS)	Berkeley, CA
<i>Student Researcher</i>	08/2023 - 12/2023
Initiated a phenomenological model to detect ultra-light dark matter from primordial black holes using asteroseismology and observations of oscillations in relativistic stars	
University of Chicago Enrico Fermi Institute	Chicago, IL
<i>UChicago Temporary Research Professional</i>	05/2023 - 08/2023
- Measured blackbody spectrum of the GigaBREAD experiment with a Vector Network Analyzer to determine system noise temperature, confirming sensitivity to $O(10)$ GHz resonant modes - Performed Fourier signal analysis of the thermal noise which identified a contaminant background	

- Wrote Python data acquisition scripts for the current best dark photon limit of 10^{-12} in [10.7, 12.5] GHz

Fermi National Accelerator Laboratory

Batavia, IL

D.O.E SULI (Science Undergraduate Laboratory Internships) Intern

01/2023 - 05/2023

- Performed optical ray-tracing simulations in FRED software to characterize focusing errors of signal photons onto a Superconducting Nanowire Single Photon Detector in InfraBREAD
- Designed an optical configuration involving a Winston Cone and parabolic reflectors that improves quantum sensing efficiency by 55%, even with misalignments from cryogenic cooling and dark matter halo velocity

Millennium Institute of Astrophysics

Santiago, Chile

Research Assistant

06/2022 - 09/2022

- Modeled the transit photometry and radial velocity data of a NASA TESS exoplanet candidate in Python, leading to the verification of the novel warm jupiter planet TOI-6628b
- Coded with Bayesian parameter optimization algorithms such as Markov Chain Monte Carlo (MCMC)

AWARDS & FELLOWSHIPS

- **NSF Graduate Research Fellowship**, National Science Foundation (2026)
- **“Outstanding Undergraduate Research Award”**, American Institute of Physics, Society of Physics Students (2024)

CONFERENCES & WORKSHOPS

- 2026 Relativity Reframed: Quantum Reference Frames and Gravity, Perimeter Institute — *poster presentation*
- 2025 & 2023 Superconducting Quantum Materials and Systems (SQMS) Center Conference, IL, — *poster presentations*
- 2025 “Quantum Sensing and Precision Science Summer School” at Johns Hopkins, MD — *participant*
- 2025 & 2024 BASICS (Bay Area Strings, Information & Cosmology Symposium), CA — *participant*
- 2025 APS Global Summit March-April Meeting, Anaheim, CA — *talk*
- 2025 “Observables in Quantum Gravity: From Theory to Experiment” Aspen Center for Physics Conference, CO — *poster presentation*
- 2024 Vienna Quantum Foundations Conference at Institute for Quantum Optics and Quantum Information (IQOQI), Vienna, Austria — *poster presentation*
- 2023 “US Quantum Information Science Summer School” at Fermilab SQMS Center, IL — *participant*

TECHNICAL SKILLS

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| <ul style="list-style-type: none"> • General research: programming, numerical simulations, data analysis • Fields: quantum information theory, quantum measurement theory, quantum reference frames, quantum optics, holographic quantum gravity • Software: Python, C++, Mathematica, Qiskit, Google Cirq, Geant4 Monte Carlo, GitHub, FRED Photon Engineering, Linux, Tex | <ul style="list-style-type: none"> • Theoretical: SYK model, JT gravity, POVMs, information scrambling, squeezed states, sub-SQL measurements, input-output theory, holographic bulk reconstruction, Euclidean and Lorentzian correlators • Experimental: superconducting qubits, beyond standard model searches, optomechanical quantum sensors, Radio-Frequency (RF) measurements, cryogenics, CCD and CMOS calorimeters |
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